Occlusion Resolver: Persistent Object Evidence and Causal Reacquisition

Research prototype and executed feasibility study

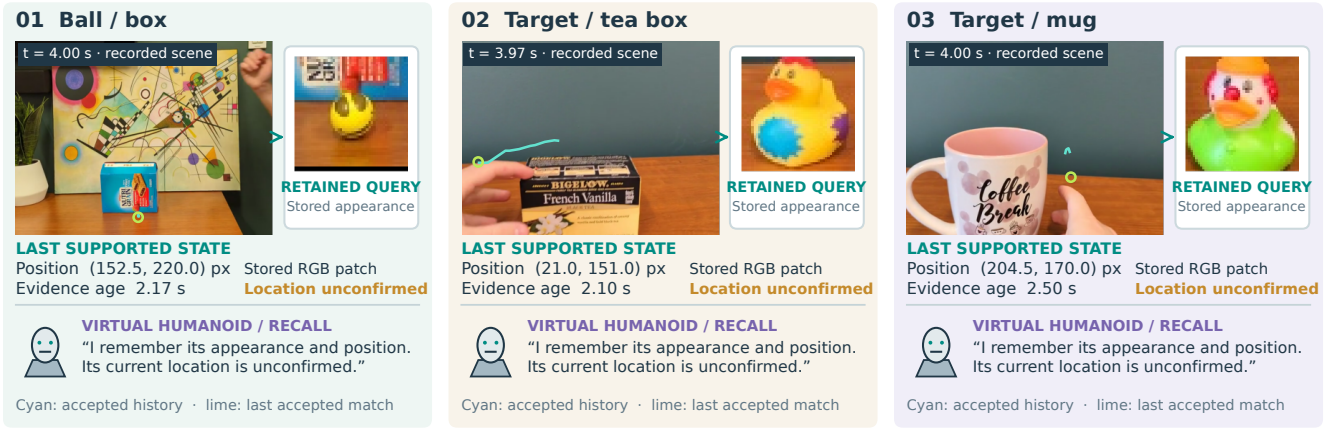


Figure 1. **Recall survives missing visual evidence.** Physical occlusion, retained query appearance, and a schematic virtual-humanoid readout. Cyan marks accepted history and lime the last accepted match, not an exact hidden location. Query crops preserve observed RGB; the dialogue is an interface template. No robot execution or semantic-property recognition is implied.

Abstract

A useful occlusion resolver must distinguish an object remembered from an object currently observed. We implement a causal system that preserves an immutable appearance query, tests reappearance candidates, retains position history when evidence is rejected, and exposes competing hidden-position hypotheses. All primary code and demonstrations concern this object-memory task. On the locally available eligible subset of TCOW Rubric Office, a new run evaluates 31 sparse target-mask endpoints from 22 clips. The resolver accepts seven target-mask hits, accepts no out-of-mask endpoints, and abstains on 24 endpoints. An appearance-only control obtains identical endpoint counts. These results establish an executable conservative prototype, not improved state of the art: confirmation has not shown an endpoint benefit, and candidate recovery is weak. We release code, traces, object-memory visualizations and a per-clip appendix so that future improvements can be tested against the same evidence contract. Exact hidden localization, learned containment, and physical humanoid transfer remain unvalidated.

1. The problem is missing evidence

When a ball moves behind a box, a current-frame detector may cease to report it. A conversational robot should not equate that absence with the claim that the ball has ceased to exist. It should also not equate the last visible coordinate with a newly measured hidden position. Those two mistakes motivate our interface: preserve the evidence, state its age, and distinguish an observation from a conditional prediction.

We focus exclusively on this problem. The earlier audio–visual switching study is not part of the present method, evaluation or contribution. The input is a video and a user-provided initial object region. The output is a persistent query record, a candidate location, an acceptance decision, and a set of explicitly unverified location hypotheses during loss. A robot-facing adapter can request a clearer view rather than inventing a confident location.

The central implementation choice is to separate identity evidence from localization state. The query RGB is immutable. Coordinates can be updated only when current evidence passes a prescribed test. This protects the identity record from being overwritten by an occluder, but it does not guarantee correct association: a visually similar distractor can still pass the test. The present architecture is intentionally inspectable, with no hidden learned module or

unreported training run.

Our release contributes an integrated implementation and an expanded diagnostic evaluation. It does not establish a new theoretical result, a learned architecture, or a top-tier novelty claim. The broader research hypothesis is that uncertainty-preserving memory can support useful reacquisition and view requests. The current endpoint results do not yet confirm an advantage over the appearance control.

2. Relation to established work

Object permanence under occlusion is not a new problem. TCOW explicitly studies targets, occluders and containers, including real-world sparse annotations [1]. Our local mask-membership evaluation is narrower than its task: it does not predict target, occluder and container segmentations or reproduce the official benchmark metric.

Point trackers such as TAPIR [2] and CoTracker3 [3] address correspondence through time. A point trajectory is not automatically a persistent semantic object record. Conversely, storing an object crop does not solve point correspondence. The website retains three archived CoTracker2/3 comparisons for inspection; these receive an offline prefix and are not compared as though they were causal online models on all 22 clips.

Memory-based segmentation systems, including XMem [4], and promptable video segmentation systems such as SAM 2 [5], provide much stronger learned representations than the classical appearance module used here. They are relevant next baselines, not methods we have evaluated in this release. Any future novelty claim must go beyond adding a memory to a tracker or attaching a robot-shaped interface to existing predictions.

3. Executed resolver

3.1. State and provenance

At initialization the query mask is converted to a bounding rectangle in a 320×240 image. The rectangle defines an RGB crop Q, a grayscale template, an HSV histogram, and an initial center. We preserve the crop exactly. Its mean RGB and dimensions are available as descriptive properties, but include crop background; they are not estimates of material, mass or object category.

The dynamic state contains the last accepted coordinate p, smoothed velocity v, elapsed time a, local-context coordinate c, and a pending candidate with a confirmation count. Every step consumes only the next decoded frame. No endpoint mask, future image, or offline tracker output enters the resolver. The point is an object-crop center, not an identified physical surface point.

3.2. Appearance proposals

For each of three scales $s \in \{0.8, 1, 1.25\}$, normalized grayscale template correlation is evaluated over the full current frame. The maximum at each scale provides one proposal. The candidate crop's HSV histogram is compared with the immutable query histogram. The combined appearance score is

$$z = 0.7\,\mathrm{NCC} + 0.3\,\mathrm{HC}, \tag{1}$$

where HC is histogram correlation. These correlations are scores, not calibrated probabilities. The highest combined score among the three proposals is retained. This search is not rotation invariant, cannot infer unseen appearance, and can miss a correct target when the strongest grayscale peak belongs to a distractor.

For ambiguity estimation we suppress a neighborhood around the peak and subtract the remaining maximum correlation from the chosen grayscale peak. A candidate qualifies when $z \geq 0.72$, $\mathrm{NCC} \geq 0.55$ and the gap is at least 0.025. These thresholds are implementation constants set before inspecting this expanded run; the data as a whole are not a pristine held-out benchmark because related examples were inspected in earlier project work.

3.3. Confirmation and state updates

The appearance control accepts any qualifying candidate. The resolver additionally requires two consecutive qualifying candidates whose displacement is less than 1.5 times the largest initial template dimension. A break in qualification resets the pending confirmation. Two similar consecutive errors can therefore still be accepted; confirmation is not an identity proof.

After acceptance, the last coordinate and local-context coordinate are set to the candidate, and evidence age is reset. Velocity is updated using the displacement since the last accepted coordinate divided by elapsed time, with equal weight on the prior velocity and the newly computed velocity. During rejection the last accepted coordinate and immutable query remain unchanged. Age increases by one frame interval. This is why the system can say what it remembers without claiming that remembered coordinates are fresh measurements.

3.4. Three conditional hypotheses

While visual support is missing, the system exposes three alternatives:

$$h_{\mathrm{hold}} = p, \tag{2}$$

$$h_{\mathrm{move}} = p + v \min(a, 1\,\mathrm{s}), \tag{3}$$

$$h_{\mathrm{context}} = c. \tag{4}$$

The first assumes the object stayed where last supported. The second extrapolates recent image motion, capped at one

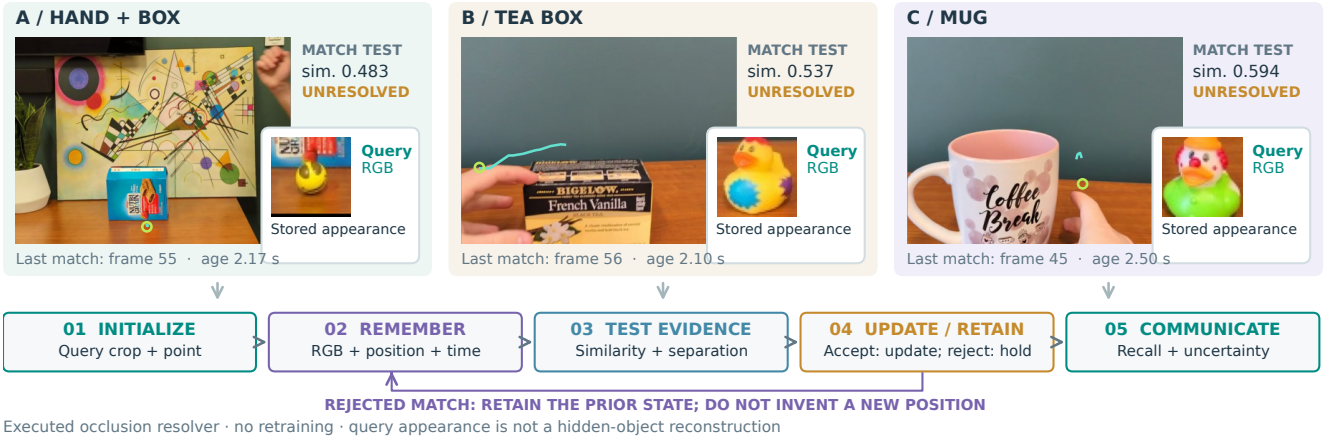


Figure 2. **Scene evidence and execution.** The gallery pairs naturally occluded scenes with remembered query appearance. The process below performs initialization, persistent storage, evidence testing, conditional update and communication. This is an implemented classical prototype; there is no claimed learned training stage.

second. The third transports a local coordinate by the median Farneback flow in a 20×20 neighborhood. Coordinates are clamped to image bounds. Context flow is a competing image-motion hypothesis, not a learned association between an object and its container.

A radius $r = 6 + 20a$ pixels is provided for display. It is a heuristic visualization, not a calibrated confidence region. The hypotheses are currently exposed to inspection and the robot-facing adapter; they do not rerank appearance proposals or improve the measured endpoint candidates. Testing their decision value is unfinished work, not an achieved contribution.

3.5. Conversational interface

The adapter returns `continue_observing` after an accepted match and `request_view_change` otherwise. The reason is explicit. A possible sentence is: "I remember the appearance and last supported position, but I cannot confirm its current location." This is a deterministic state readout. No language model, Ameca SDK, gaze actuator, or physical robot is part of the executed experiment.

4. Evaluation contract

We enumerate locally available videos with an initial query mask and a later `snitch` target mask. Of 37 local videos, 22 satisfy the evaluated conditions, yielding 31 annotated endpoints. We initialize from the earliest query mask and process every frame through the last scored endpoint. The query mask determines initialization only. Later masks are opened only in the evaluation stage after predictions are computed.

A candidate is a hit if its scaled, rounded coordinate lies inside the provided target mask. We report candidate hits, accepted hits, accepted misses and abstentions. Mask membership does not establish exact point correspondence, correct identity throughout the interval, or accurate localization during total invisibility. We do not infer dense visibility labels from these sparse masks. Some clips contribute multiple endpoints, so endpoints are not independent trials.

Method	Candidate hits	Accept hits	Wrong accept	Abstain
Resolver	9	7	0	24
Appearance	9	7	0	24
Stationary	7	7	24	0

Table 1. 31 sparse endpoints from 22 clips. Counts, not percentages. Candidate hits ignore rejection. The resolver and appearance control tie on all aggregate counts.

The expanded controls are: (1) the same appearance proposals and thresholds without the two-frame confirmation, and (2) a stationary initial center accepted at every endpoint. Both use the same query bounding rectangle. These controls isolate the confirmation effect and the effect of refusing to answer. They do not constitute a comprehensive comparison against learned state-of-the-art systems.

We release the complete per-endpoint table, not selected successes. The visualization clips are inspected demonstrations and are not a random sample. All constants, resize conventions and initialization rules are released. No tuning sweep or model selection based on this run was performed. Timing is local wall time for evaluation code, not an optimized robot-latency benchmark.

5. Results and what they do not show

The resolver accepts seven target-mask hits and no out-of-mask endpoints. Its acceptance rate is $7/31 = 22.6\%$, leav-

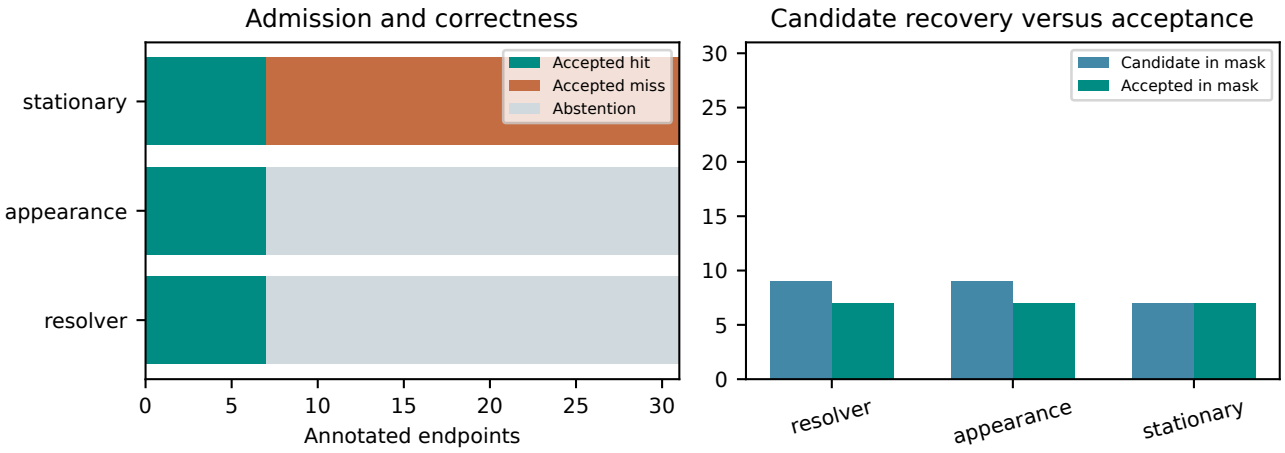


Figure 3. **Conservatism is not recovery.** The resolver avoids accepted misses in this inspected endpoint set by rejecting many cases. Its candidate generator produces nine hits and its acceptance stage retains seven. Confirmation does not improve the endpoint counts over appearance alone.

ing 77.4% of the endpoints unanswered. Its unconditional candidate lies in the mask at nine endpoints. Thus two candidate hits are rejected, while most candidate proposals are already incorrect before the acceptance decision is applied.

The confirmation rule obtains exactly the same aggregate counts as the appearance control. This is a negative result for the proposed advantage of adding confirmation at the scored endpoints. It does not rule out differences at intermediate frames, but no dense labels exist in this evaluation to establish that those differences would be beneficial.

The stationary control has seven hits and 24 misses. Comparing accepted errors alone would make the resolver look much stronger while hiding how often it declines to answer. Comparing accepted hits reveals that it does not increase the total number of correct supported answers in this run. The two controls can succeed on different cases; we therefore release per-endpoint results rather than substituting equal totals for a paired analysis.

Zero accepted misses is not evidence of a zero population error rate. There are only seven accepted endpoints, clips are clustered, the dataset is narrow, and the development history is inspected. We do not call this a calibrated safety guarantee. The expanded execution took approximately 64 seconds on the local environment, including decoding and both appearance variants. This number is not comparable with published GPU throughput.

6. Failure analysis

Appearance mismatch. Changes in viewpoint, illumination, scale and orientation can destroy correlation with an immutable query. Three scales do not cover rotations or substantial deformations. Maintaining an immutable identity anchor avoids destructive overwrites but makes the representation less adaptable.

Occluder confusion. Context flow follows local texture. That texture may belong to the foreground occluder, background or hand. Treating it as certain target motion would convert an explicit hypothesis into an unsupported hidden-location claim. We expose it separately for this reason.

Missed reacquisition. Requiring consecutive evidence can delay reacquisition, and the ambiguity threshold can reject a correct peak. At the current endpoints the control ties the resolver, so confirmation is not justified as a measured improvement. A learned re-identification representation or a discriminative appearance bank is a plausible next intervention, but neither is implemented or evaluated here.

Semantic overstatement. An RGB crop preserves color and texture measurements; it does not establish that an object is a yellow rubber toy, its affordance, or its material. The demonstration labels refer to the scene content as described by a human. The code exports observed properties and does not pretend to have inferred physical properties.

7. Reproduction and next decisive tests

Run `code/evaluate.py` with the local Rubric Office directory and an output directory. The script opens the earliest query annotation, executes causal frame updates and then evaluates later masks. `code/test_resolver.py` checks immutable memory during rejection, two-frame confirmation, prefix determinism, and finite separate hypotheses. All four implementation tests pass. These tests protect

code behaviour; they do not substitute for accuracy evaluation.

The next useful experiment should test a mechanism, not just add an impressive visualization. A motion-hypothesis ablation must actually let hypotheses influence candidate selection and compare it against appearance alone on a locked split. A containment-aware model must predict object–container association from pixels and be evaluated on hidden movement and reveal events. An active-view policy must receive alternative views or real camera control and measure reduction in localization uncertainty. The present single-camera recordings cannot establish that a proposed view request would reveal the target.

The project is now organized around occlusion resolution, but the current method is a conservative classical prototype. A credible novel method paper still requires stronger learned baselines under matched information access, broader datasets, explicit hidden-state annotations, calibrated uncertainty, and a demonstrated gain from the proposed mechanism. These are outstanding research requirements, not claims fulfilled by this release.

References

[1] B. Van Hoorick et al. Tracking Through Containers and Occluders in the Wild. CVPR, 2023. https://tcow.cs.columbia.edu/.

[2] C. Doersch et al. TAPIR: Tracking Any Point with Per-Frame Initialization and Temporal Refinement. ICCV, 2023. https://arxiv.org/abs/2306.08637.

[3] N. Karaev et al. CoTracker3: Simpler and Better Point Tracking by Pseudo-Labelling Real Videos. ICCV, 2025. https://arxiv.org/abs/2410.11831.

[4] H. K. Cheng and A. G. Schwing. XMem: Long-Term Video Object Segmentation with an Atkinson–Shiffrin Memory Model. ECCV, 2022. https://arxiv.org/abs/2207.07115.

[5] N. Ravi et al. SAM 2: Segment Anything in Images and Videos. arXiv:2408.00714, 2024. https://arxiv.org/abs/2408.00714.